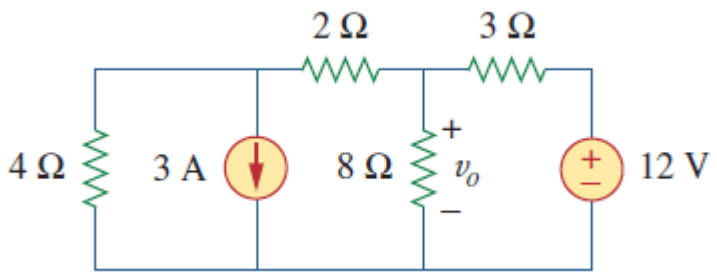
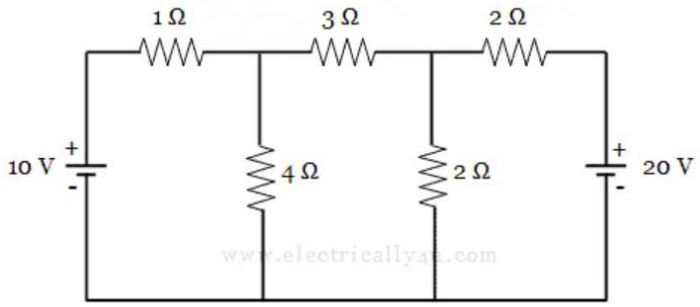
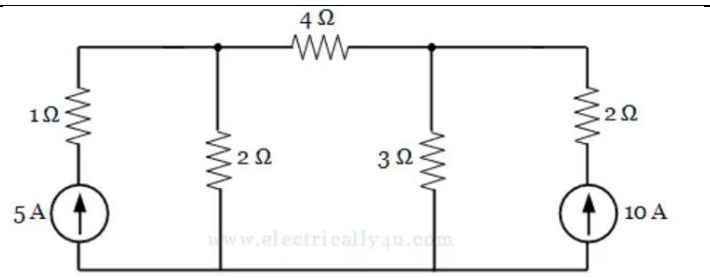
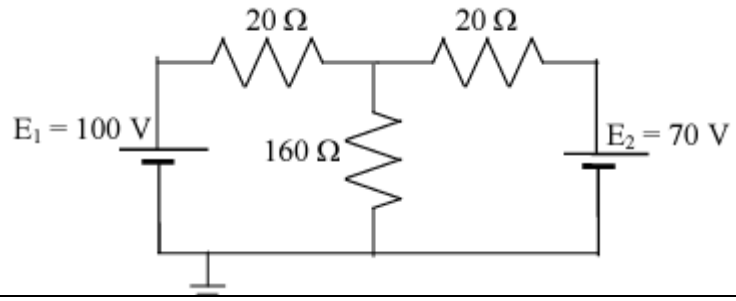
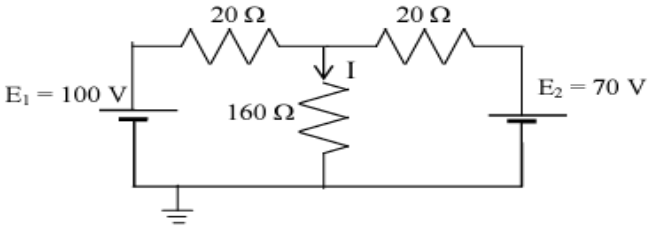
Class : I B. Tech – II Sem
Course : Network Analysis
Faculty : Dr. Prabhuraj S / R. Yedukondalu

Academic Year : 2023-2024
Branch : ECE

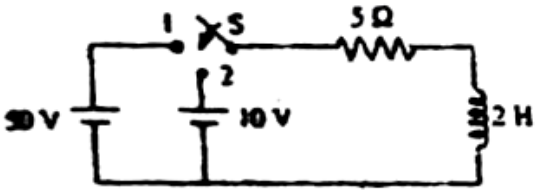
MID QUESTION BANK

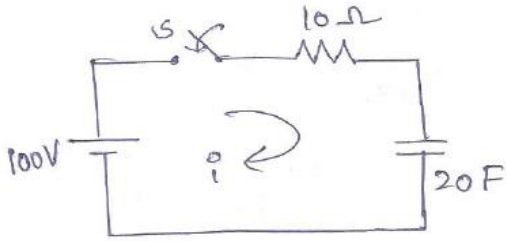
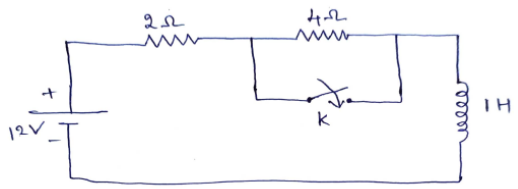
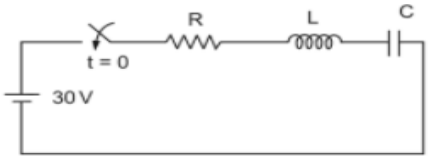
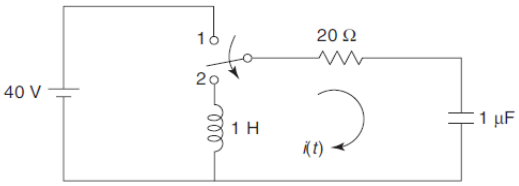
UNIT-I

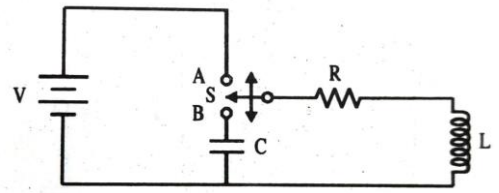
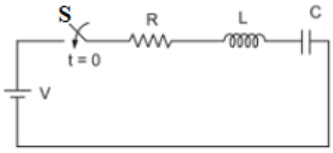
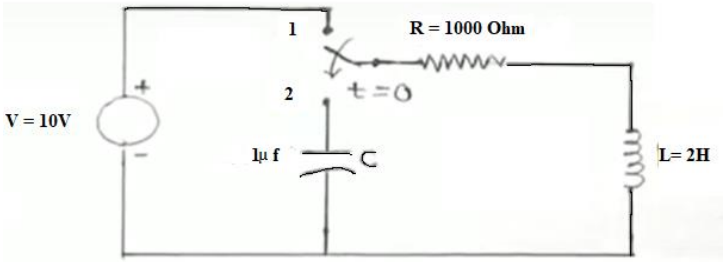
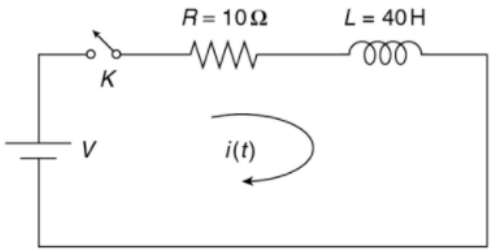
S.NO	QUESTION	CO	BL	MARKS
1	Explain the types of circuit components,	1	2	10
2	Write the Types of Sources and Source Transformations,	1	2	10
3	Explain the mesh analysis with example.	1	2	10
4	Explain the nodal analysis with example.	1	2	10
5	Use source transformation to find V0 in the circuit of Fig. 	1	2	10
6	State and explain of super position theorem with example.	1	2	10
7	State and explain reciprocity theorem with example.	1	2	10
8	Explain the principle of Duality with an example	1	2	10
9	Determine the mesh currents in the circuit shown below. 	1	2	10
10	Determine the current through each resistor in the given circuit using nodal analysis.	1	2	10

				
11	Find the current using thevenins theorem 	1	2	10
12	The figure shown (also used in earlier examples) can be considered equivalent to two sources of 100 V and 70 V, with internal resistances 20 Ω each, feeding a load of 160 Ω. Using Millmann's theorem (or equivalent generator theorem) find the current I. 	1	2	10

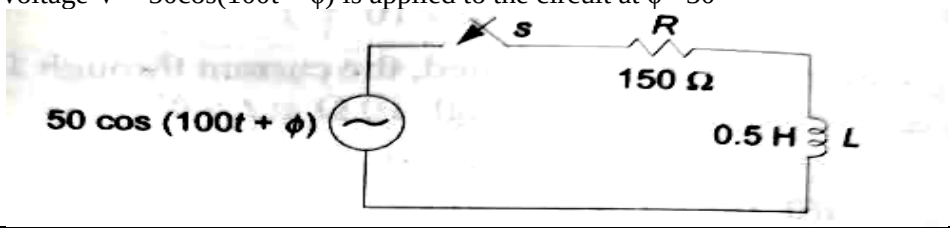
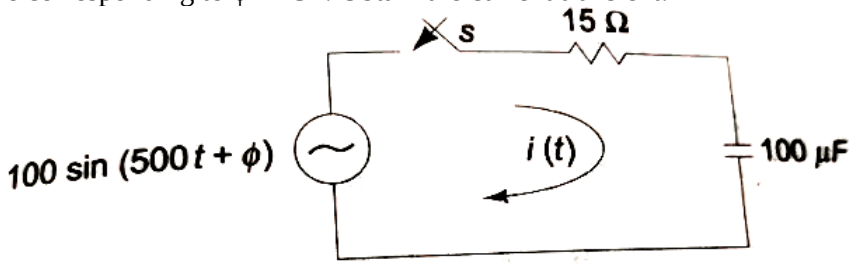
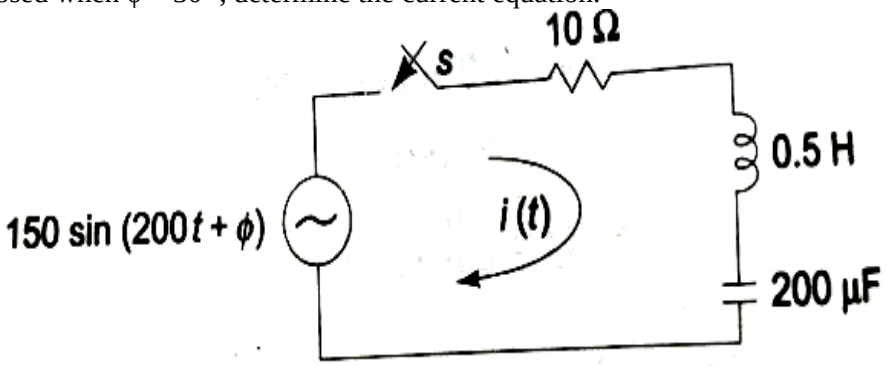
UNIT-II

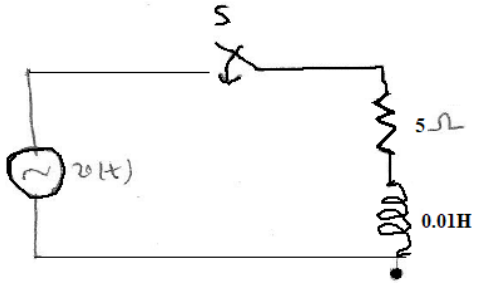
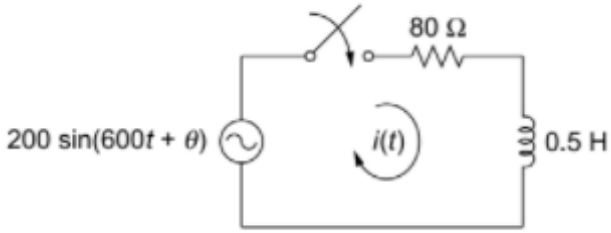
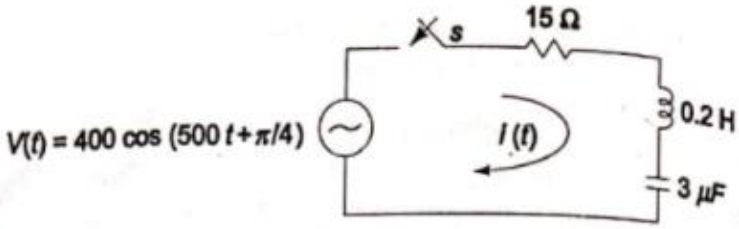
S.NO	QUESTION	CO	BL	MARKS
1	Derive an expression for i(t) of a RL series circuit when DC voltage is applied at t=0 by closing the switch. Draw the response curve i(t). Define time constant of RL circuit	2	2	10
2	A series RL circuit with R= 30 Ohm, L= 15 H, has a constant voltage V = 60V applied at t= 0. Determine the current I and the voltage across the resistor and voltage across the Inductor.	2	2	10
3	In the circuit as shown in figure below the switch is in position1 for along time and brought the position2 at time t=0. Determine the circuit current. 	2	3	10
4	Derive an expression for i(t) of a RC series circuit when DC voltage is applied at t=0 by closing the switch. Draw the response curve i(t). Define time constant of RC circuit.	2	2	10

5	A series RC circuit with R = 10ohm and C= 20F has a constant voltage of V = 100V applied at t=0. Determine the current i and the voltage across the resistor and voltage across the capacitor.  Figure(a)	2	2	10
6	For the figure(a) shown in below, the dc voltage is applied to the circuit keeping the switch K open so that the steady state is reached. Determine the complete response for the circuit after closing the switch K.  Figure(a)	2	2	10
7	Consider series RLC circuit as shown in below figure (a) with R= 2**Ω**, L=0.5H and C=0.1F.the circuit is initially relaxed. When the switch is closed at t=0. Calculate the transient current and voltage across the capacitor.  	2	2	10
8	In the network shown in below Figure (a) the switch is changed from position 1 to the position 2 at t=0, steady condition having reached before switching. Find the values of I, di/dt and d^2i/dt^2at t= 0+.  Figure (a)	2	2	10
9	In the network shown in below figure (a) the switch is changed from position A to the position B at t=0, having already established steady state in position A. Find the values of i, di/dt and d^2i/dt^2 at t= 0+.If R=100**Ω**, L=1H and C= 1μF and V= 100 volt.	2	2	10

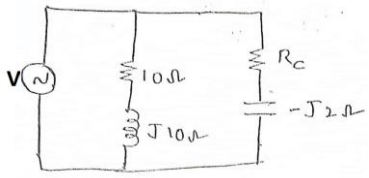
	 Figure (a)			
10	In the network shown in below figure (a) the switch S is closed at t= 0 with zero inductor current and capacitor uncharged. find i, di/dt and d^2i/dt^2 at t= 0+.If R=10**Ω**, L=1H and C= 100μF and V= 100 volt.  Figure (a)	2	2	10
11	The switch in below figure(a) is changed from position 1 to 2 at t=0. find i, di/dt and d^2i/dt^2 at t= 0+.  Figure (a)	2	2	10
12	The values of R and L in a series R-L circuit are 10**Ω** and 40 H, respectively. At the instant of closing the switch, the current rises at the rate of 5A/s. Calculate the following:1. the value of applied voltage. 2. Rate of growth of current when 6A flows in the circuit3. Find the energy stored in the inductor.  Figure(a)	2	2	10

UNIT-III

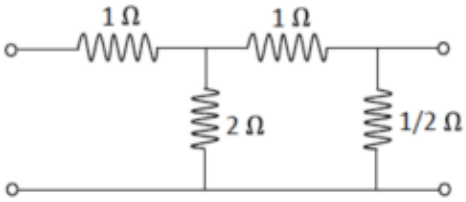
S.NO	QUESTION	CO	BL	MARKS
1	a) What is transient state?	3	2	05
	b) Why transients occurs in electric circuits.	3	2	05
2	Derive an expression for transient response in series RC circuit with sinusoidal excitation	3	3	10
3	Derive an expression for transient response in series RL circuit with sinusoidal excitation	3	3	10
4	The circuit shown in fig, consists of series RL elements with R = 150Ω and L = 0.5 H. the switch is closed when $\phi = 30^0$. determine the resultant current when voltage V = 50cos(100t + ϕ) is applied to the circuit at $\phi = 30^0$ 	3	3	10
4	The circuit shown in fig , consists of series RC elements with R = 15Ω and C = 100μF. A sinusoidal voltage v = 100 sin(500t + ϕ) volts is applied to the circuit at time corresponding to $\phi = 45^0$. Obtain the current transient. 	3	3	10
5	Explain the AC response for RL circuits by using solution method of differential equations	3	2	10
6	Explain the AC response for RLC circuits by using solution method of differential equations	3	2	10
7	The circuit shown in fig. consisting of series RLC elements with R = 10Ω, L= 0.5 H and C = 200μF has a sinusoidal voltage v = 150 sin(200t + ϕ). If the switch is closed when $\phi = 30^0$, determine the current equation. 	3	2	10
8	In the series RL circuit shown in below figure (a) the sinusoidal source is given by V= 100sin(500t+φ) volts. Determine resetting current if the switch is closed when φ=0.	3	2	10

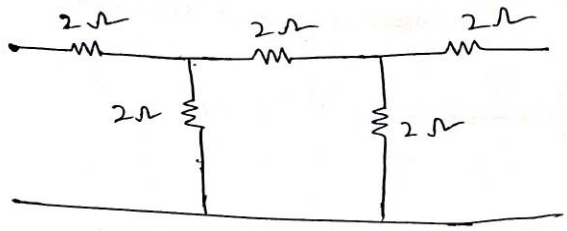
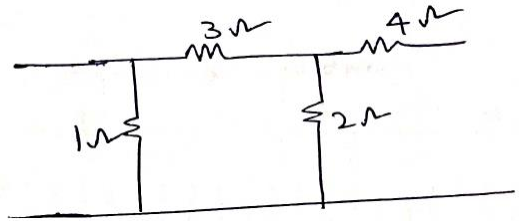
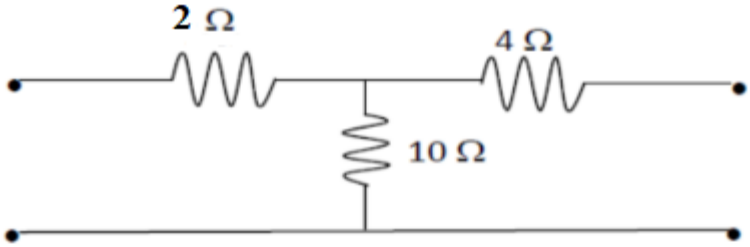
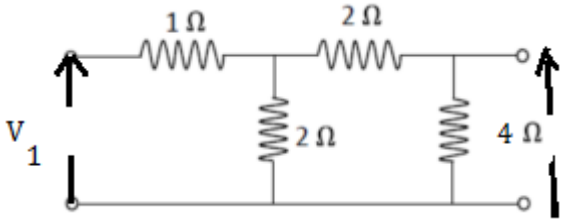
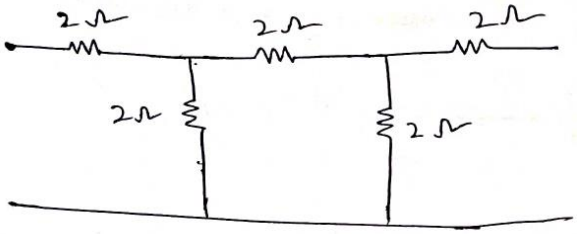
	 Figure (a)			
9	A series R-C circuit has R=20**Ω** and C= 100μF. A voltage of *v*= 200 sin 314t is applied at t=2.14msec. Obtain an expression for *i*. Also find the value of current after 1m sec from the switching instant.	3	2	10
10	A 50Hz, 400V(peak value) sinusoidal voltage is applied at t= 0 to 2πf, series R-L circuit having resistance 5**Ω** and inductance 0.2H. Obtain an expression of current at any instant t. calculate the value of the transient current at 0.01 sec after switching on.	3	2	10
11	For the network shown in below figure (a), a sinusoidal voltage source v= 200 sin (600t+θ)volt is applied at a time when θ=0°. find the expression for current i(t)?  Figure (a)	3	2	10
12	In the circuit shown in below figure (a) determine the complete solution for the current, when the switch is closed at t=0. Applied voltage v(t)=400cos(500t+π/4). Resistance R=15**Ω** Inductance L= 0.2H and capacitance C=3μF.  Figure (a)	3	2	10

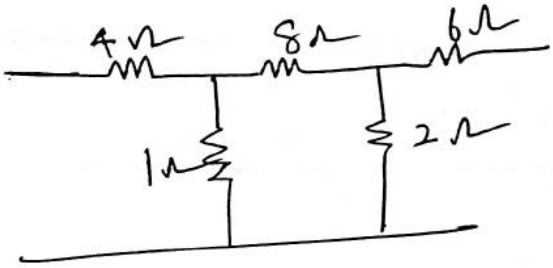
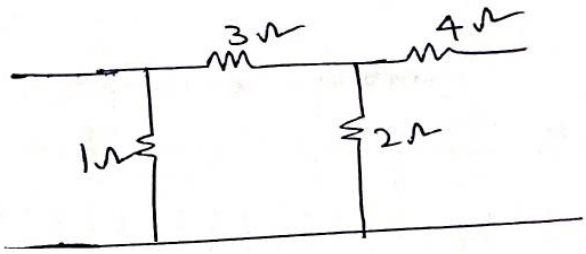
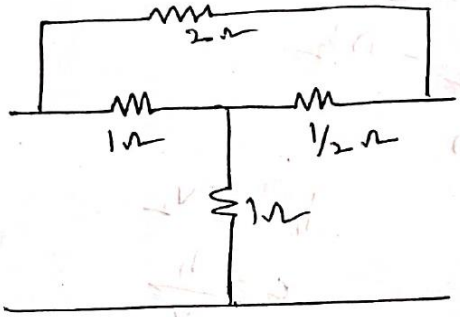
UNIT-IV

S.NO	QUESTION	CO	BL	MARKS
1	Derive the Quality factor of the series RLC circuit at Resonance. Also derive the resonant frequency, quality factor and bandwidth.	4	2	10
2	Derive the Quality factor of the Parallel RLC circuit at Resonance. Also derive the resonant frequency, quality factor and bandwidth	4	2	10
3	a) Explain the Dot rule of Coupled Circuits	4	2	10
	b) Explain the Self Inductance and Mutual Inductance	4	2	10
4	Derive the expression for Co-efficient of Coupling	4	2	10
5	Explain the analysis of coupled circuits	4	2	10
6	Explain the Conductively coupled equivalence circuits	4	2	10
7	For a series RLC circuit, R = 10 ohms, L = 0.3H and C = 1006 μF. The applied voltage is 230V. Determine resonance frequency, quality factor, current at resonance, voltage across inductance at resonance, bandwidth and half power frequencies.	4	2	10
8	Determine the value of R_c for which the network shown in figure at Resonance? 	4	2	10
9	For a series RLC circuit, R = 25 ohms, L = 10mH and C = 0.06 μF. Find the value of Bandwidth and Q factor.	4	2	10
10	A coil of resistance 20 ohm and inductance 10mH in series with a capacitance and is supplied with a constant voltage, variable frequency source. The maximum current is 2A at 1000Hz. Find the cutoff frequencies.	4	2	10

UNIT-V

S.No	Question	CO	BL	MARKS
1.	Find the Y-parameters for the network shown in below figure (a)  Figure (a)	5	2	10
2.	a) Define and explain h- parameters of a two port network	5	2	5
	b) Derive an expression for z parameters in terms of a y parameters.	5	2	5

3	Derive and find the h-parameters for the network shown in figure (a)  Figure (a)	5	2	12
4	A) Obtain Y Parameters for the network shown in below figure (a)  Figure (a)	5	2	12
5	(a) Find the h-parameters for the network shown in figure below figure (a)  Figure (a)	5	2	5
	(b) What is two port networks? Briefly explain with the help of a block diagram.	5	2	5
6.	Find the Z parameters for the network shown in figure below figure (a)  Figure (a)	5	2	10
7.	Find the transmission parameters for the network shown in figure below figure (a)  figure (a)	5	2	10

8.	Derive an expression for z parameters in terms of h-parameter	5	2	5
	Derive an expression for y parameters in terms of h parameters	5	2	5
9	For the network shown in below figure (a) find the ABCD Parameters  Figure (a)	2	10
10	Derive the ABCD parameters in terms of Z parameters and Y parameters	5	2	10
11	Determine the h – parameters for the network shown in below figure (a)  Figure (a)	5	2	10
12	Determine the transmission parameters for the network shown in below figure (a)  Figure (a)	5	2	10

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